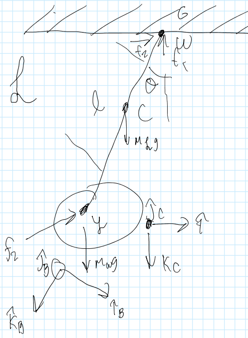
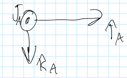
Dynamics problem

Wednesday, August 13, 2025 11:04 AM





$F_A \xrightarrow[2]{\theta} F_B$

$F_A \xrightarrow[2]{\phi} F_C$

$r_{C/W} = \frac{\ell}{2}\hat{k}_B$

Find $\ddot{\theta}, \ddot{\phi}$

$$O_{B/A} = O_2(\theta) = {}_B\begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}^A$$

$\omega_{B/A} = \dot{\theta}\,\hat{j}_A \qquad \omega_{B/A} = \dot{\theta}\,\hat{j}_A = \dot{\theta}\hat{j}_B$

$\omega_{C/A} = \dot{\phi}\,\hat{j}_A \qquad \overset{A}{\dot{\omega}}_{C/A} = \ddot{\phi}\hat{j}_A = \ddot{\phi}\hat{j}_B$

$\vec{v}_{c/w/A} = \frac{\ell}{2}\overset{A}{\dot{k}}_B = \frac{\ell}{2}(\omega_{B/A} \times k_B) = \frac{\ell}{2}(\dot{\theta}\hat{j}_A \times k_B) = \frac{\ell}{2}(\dot{\theta}\hat{j}_B \times k_B) = \frac{\ell}{2}\dot{\theta}\hat{i}_B$

$a_{c/w/A} = \frac{\ell}{2}\ddot{\theta}\hat{i}_B + \frac{\ell}{2}\dot{\theta}\overset{A}{\dot{\hat{i}}}_B = \frac{\ell}{2}(\ddot{\theta}\hat{i}_B + \dot{\theta}(\omega_{B/A} \times \hat{i}_B)) = \frac{\ell}{2}(\ddot{\theta}\hat{i}_B + \dot{\theta}(\dot{\theta}\hat{j}_B \times \hat{i}_B)) = \frac{\ell}{2}(\ddot{\theta}\hat{i}_B - \dot{\theta}^2 k_B)$

$\vec{f}_{\mathcal{L}} = M_{\mathcal{L}} g\,\hat{k}_A + \vec{f}_{r/\mathcal{L}/G} + f_{r/\mathcal{L}/W} = M\,\vec{a}_{c/w/A}$

$\vec{r}_{Y/W} = \ell k_B$

$\vec{v}_{Y/W/A} = \overset{A}{\dot{\vec{r}}}_{Y/W} = \ell\overset{A}{\dot{k}}_B = \ell(\omega_{B/A} \times k_B) = \ell\dot{\theta}\hat{i}_B$

$M_W a_{Y/W/A} = f_w = M_W\vec{g} + \vec{f}_{r/w/\mathcal{L}} = M_W\vec{g} - \vec{f}_{r/\mathcal{L}/w}$

$a_{Y/W/A} = \overset{A}{\ddot{r}}_{Y/W} = \ell\overset{A}{\ddot{k}}_B = \ell\ddot{\theta}\hat{i}_B + \ell\dot{\theta}\overset{A}{\dot{\hat{i}}}_B = \ell(\ddot{\theta}\hat{i}_B - \dot{\theta}^2 k_B)$

$f_{r/\mathcal{L}/w} = M_W a_{Y/W/A} - M_W g$

$= M_W(\ell\ddot{\theta}\hat{i}_B - \dot{\theta}^2 k_B) - g)$

$\frac{M\ell}{2}(\ddot{\theta}\hat{i}_B - \dot{\theta}^2 k_B) = Mg k_A + f_{r/\mathcal{L}/G} + M_W(\ell\ddot{\theta}\hat{i}_B - \dot{\theta}^2 k_B - g(-\sin\theta\hat{i}_B + \cos\theta\, k_B)$

$\frac{M\ell}{2}(\ddot{\theta}\hat{i}_B - \dot{\theta}^2 k_B) - Mg(-\sin\theta\hat{i}_B + \cos\theta k_B) - M_W(\ell\ddot{\theta}\hat{i}_B - \dot{\theta}^2 k_B - g(-\sin\theta\hat{i}_B + \cos\theta k_B) = F_{r/\mathcal{L}/G}$

$\vec{f}_{\mathcal{L}} = \vec{f}_{r/\mathcal{L}/G} + M\vec{g} + f_{r/\mathcal{L}/w}$

$= \frac{M\ell}{2}(\ddot{\theta}\hat{i}_B - \dot{\theta}^2 k_B) - \cancel{Mg(-\sin\theta\hat{i}_B + \cos\theta k_B) - M_W(\ell\ddot{\theta}\hat{i}_B - \dot{\theta}^2 k_B - g)} + \cancel{Mg k_A} + \cancel{M_W((\ell\ddot{\theta}\hat{i}_B - \dot{\theta}^2 k_B) - g)}$ $\cancel{(-\sin\theta\hat{i}_B + \cos\theta k_B)}$

$\frac{1}{4} + \frac{1}{12} = \frac{3}{12} + \frac{1}{12} = \frac{4}{12} = \frac{1}{3}$

$J_{\mathcal{L}/C} = \frac{M\ell^2}{12}\begin{bmatrix} 1&0&0 \\ 0&1&0 \\ 0&0&0 \end{bmatrix}_B = \frac{M\ell^2}{12}(\hat{i}_B\hat{i}_B' + \hat{j}_B\hat{j}_B')$

$r_{W/C} = -\frac{\ell}{2}k_B$

$\vec{J}_{\mathcal{L}/W} = \vec{J}_{\mathcal{L}/C} + M(|r_{W/C}|^2 I - r_{W/C}r_{W/C}')$

$= \frac{M\ell^2}{12}(\hat{i}_B\hat{i}_B' + \hat{j}_B\hat{j}_B') + M(\frac{\ell^2}{4}(\hat{i}_B\hat{i}_B' + \hat{j}_B\hat{j}_B' + \cancel{k_B k_B'}) - \cancel{(\frac{\ell^2}{4}k_B k_B')})$

$= \frac{M\ell^2}{12}(\hat{i}_B\hat{i}_B' + \hat{j}_B\hat{j}_B') + \frac{M\ell^2}{4}(\hat{i}_B\hat{i}_B' + \hat{j}_B\hat{j}_B')$

$= \frac{M\ell^2}{3}(\hat{i}_B\hat{i}_B' + \hat{j}_B\hat{j}_B')$

$\ell k_B \times (\frac{M\ell}{2}\ddot{\theta} + Mg\sin\theta - M_W\ell\ddot{\theta} + g\sin\theta)\hat{i}_B + (-\frac{M\ell}{2}\dot{\theta}^2 - Mg\cos\theta + M_W\dot{\theta}^2 - g\cos\theta)k_B$

$= \ell k_B \times (M(\frac{\ell}{2}\ddot{\theta} + g\sin\theta(1 + \frac{1}{M})) - M_W\ell\ddot{\theta})\hat{i}_B + M(-\frac{\ell}{2}\dot{\theta}^2 - g\cos\theta(1 + \frac{1}{M})) + M_W\dot{\theta}^2)k_B$

A $\hat{i}_B$ B k_B

$= M(\frac{\ell}{2}\ddot{\theta} + g\sin\theta(1 + \frac{1}{M})) - M_W\ell\ddot{\theta})\hat{j}_B$

$\begin{bmatrix} \hat{i} & \hat{j} & k \\ c & 0 & \frac{\ell}{2} \\ -Mg\sin\theta & 0 & Mg\cos\theta \end{bmatrix} = 0\hat{i}_B + 0k_B + -\frac{\ell}{2}Mg\sin\theta\hat{j}_B$

$\begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & \ell \\ A & 0 & B \end{bmatrix} = 0\hat{i}_B + A\ell\hat{j}_B + 0\hat{k}_B$

$M_{\mathcal{L}/W} = \vec{J}_{\mathcal{L}/W}\overset{A}{\dot{\omega}}_{B/A}$

$= \left(\frac{M\ell^2}{3}(\hat{i}_B\hat{i}_B' + \hat{j}_B\hat{j}_B')\right)(\ddot{\theta}\hat{j}_B)$

$= \vec{r}_{C/W} \times M\vec{g} + r_{Y/W} \times f_{r/\mathcal{L}/W} + \tau_{r/\mathcal{L}/W}$

$= \frac{\ell}{2}k_b \times Mg\cancel{k_A} + \ell k_b \times (\frac{M\ell}{2}(\ddot{\theta}\hat{i}_b - \dot{\theta}^2 k_b) - Mg(-\sin\theta\hat{i}_b + \cos\theta k_B) - M_W(\ell\ddot{\theta}\hat{i}_B - \dot{\theta}^2 k_b - g(-\sin\theta\hat{i}_B + \cos\theta k_b)) + \tau_{r/\mathcal{L}/W}$

$(-\sin\theta\hat{i}_b + \cos\theta k_B)$

$= -\frac{\ell}{2}\sin\theta\hat{j}_B + (M\frac{\ell}{2}\ddot{\theta} + Mg\sin\theta + g\sin\theta - M_W\ell\ddot{\theta})\hat{j}_B + \tau_{r/\mathcal{L}/W}$

$\tau_{r/\mathcal{L}/W} = \frac{\ddot{\theta}M\ell}{3} + \frac{\ell}{2}\sin\theta - M\frac{\ell}{2}\ddot{\theta} - Mg\sin\theta - g\sin\theta + M_W\ell\ddot{\theta}$

$= M(\frac{\ddot{\theta}\ell}{3} + \frac{\ddot{\theta}\ell}{2} - g\sin\theta(1 + \frac{1}{M})) + \frac{\ell}{2}\sin\theta + M_W\ell\ddot{\theta}$

$= M(\frac{5\ddot{\theta}\ell}{6} - g\sin\theta(1 + \frac{1}{M})) + \frac{\ell}{2}\sin\theta + M_W\ell\ddot{\theta})$

$\vec{M}_{W/Y} = -\vec{T}_{r/\mathcal{L}/W}$

$= J_{W/Y}\overset{A}{\dot{\omega}}_{C/A}$

$= \frac{Mr^2}{4}\begin{bmatrix} 1&0&0 \\ 0&2&0 \\ 0&0&1 \end{bmatrix} \cdot \ddot{\phi}\hat{j}_B$

$= \frac{2\ddot{\phi}Mr^2}{4}\hat{j}_B = \frac{\ddot{\phi}Mr^2}{2}$

$J_{W/Y} = \frac{Mr^2}{4}\begin{bmatrix} 1&0&0 \\ 0&2&0 \\ 0&0&1 \end{bmatrix} = \frac{Mr^2}{4}[\hat{i}_B\hat{i}_B' + 2\hat{j}_B\hat{j}_B' + k_B k_B']$

$\frac{\ddot{\phi}\cancel{M}r^2}{2} = -\cancel{M}(\frac{5\ddot{\theta}\ell}{6} - g\sin\theta(1 + \frac{1}{M})) + \frac{\ell}{2}\sin\theta + M_W\ell\ddot{\theta})$

$\ddot{\phi} = -\frac{2}{r^2}(\frac{5\ddot{\theta}\ell}{6} - g\sin\theta(1 + \frac{1}{M}) + \frac{\ell}{2M}\sin\theta + \frac{M_W}{M}\ell\ddot{\theta})$

$$\dot{\phi} = \frac{-2}{r^2}\left(\frac{5\ddot{\theta}l}{6} - g\sin\theta\left(1+\frac{1}{M}\right) + \frac{l}{2M}\sin\theta + \frac{M_w}{M}l\ddot{\theta}\right)$$